



POLITECNICO
MILANO 1863

SPACECRAFT ATTITUDE DYNAMICS AND CONTROL

Lab 6 – Gravity Gradient perturbation

PROF. FRANCO BERNELLI

PIETRO BRUSCHI, MASSIMILIANO BUSSOLINO, ANNACHIARA IPPOLITO, ANDREA COLAGROSSI

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Second Tutoring session

Friday 24 October

Room L.11

14:15 – 16:15

Due to Wednesday 22nd - Write questions to me: annachiara.ippolito@polimi.it

These hours are meant to fill the gap on the fundamentals of MATLAB and Simulink.

Partecipate if you have questions: it will not be a lecture but an interactive session!

www.tutorapp.polimi.it -> try doing a first access!

The day of the tutoring you must register **within the first half an hour.**

My person code: 10661425

Task 1: Compute the relative attitude between the S/C body frame and a moving reference frame

Principal moments of Inertia

$$I_x = \{0.040, 0.060\} \text{ kg m}^2$$

$$I_y = \{0.060, 0.080\} \text{ kg m}^2$$

$$I_z = \{0.080, 0.040\} \text{ kg m}^2$$

Initial conditions

$$\omega_x(0) = \{0, 10^{-6}\} \text{ rad/s}$$

$$\omega_y(0) = \{0, 10^{-6}\} \text{ rad/s}$$

$$\omega_z(0) = n \text{ rad/s}$$

$$A_{B|N}(0) = A_{L|N}(0)$$

Governing Equations

Dynamics

$$\begin{cases} \dot{\omega}_x = \frac{I_y - I_z}{I_x} \omega_y \omega_z \\ \dot{\omega}_y = \frac{I_z - I_x}{I_y} \omega_x \omega_z \\ \dot{\omega}_z = \frac{I_x - I_y}{I_z} \omega_y \omega_x \end{cases}$$

Kinematics

$$\frac{dA_{B|N}(t)}{dt} = -[\omega]^\wedge A(t)$$

Reference Attitude

$$A_{L|N} = \begin{bmatrix} \cos(nt) & \sin(nt) & 0 \\ -\sin(nt) & \cos(nt) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boldsymbol{\omega}_{L|N} = \begin{bmatrix} 0 \\ 0 \\ n \end{bmatrix}$$

$$\text{with } n^2 = \frac{GM_t}{R^3}$$

You have a spacecraft orbiting Earth at a certain altitude H . You want the spacecraft to follow the reference moving attitude given by $A_{L|N}$ and $\boldsymbol{\omega}_{L|N}$, but currently you have no controls on your satellite.

Tasks

- Compute the attitude error between the S/C orientation and the reference orientation at each timestep.

Attitude error

$$A_{B|L} = A_{B|N} A_{L|N}^T$$

Ang. vel. error

$$\boldsymbol{\omega}_{B|L} = \boldsymbol{\omega}_B - A_{B|L} \boldsymbol{\omega}_L$$

Task 2: Repeat including the Gravity Gradient perturbation

Governing Equations

Dynamics

$$\begin{cases} \dot{\omega}_x = \frac{I_y - I_z}{I_x} \omega_y \omega_z - \frac{3GM_t}{R^3} \frac{I_y - I_z}{I_x} c_2 c_3 \\ \dot{\omega}_y = \frac{I_z - I_x}{I_y} \omega_x \omega_z - \frac{3GM_t}{R^3} \frac{I_z - I_x}{I_y} c_1 c_3 \\ \dot{\omega}_z = \frac{I_x - I_y}{I_z} \omega_y \omega_x - \frac{3GM_t}{R^3} \frac{I_x - I_y}{I_z} c_2 c_1 \end{cases}$$

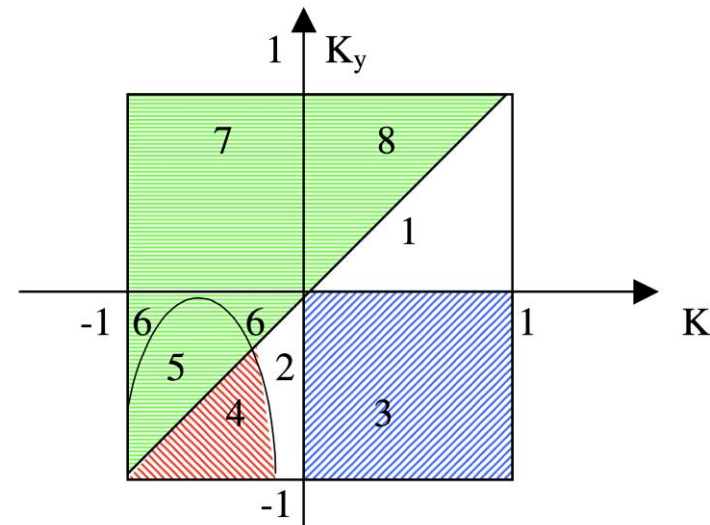
with

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = A_{B|L} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

(try to use vectorial form!)

Tasks

- Simulate the spacecraft dynamics and kinematics under the effect of the gravity gradient perturbation.
- Compute the attitude error with respect to the reference attitude
- Validate some points in the stability region diagram



$$\begin{aligned} K_x &= \frac{I_z - I_y}{I_x} \\ K_y &= \frac{I_z - I_x}{I_y} \\ K_z &= \frac{I_y - I_x}{I_z} \end{aligned}$$